Department of Computer Science Human Computer Interaction Course COMP322

“QOU Student Portal: Schedule, Grades and Courses Tracking Redesign using UCD”

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Section 1

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QOU Student Portal Redesign — Design Rationale:

1. Conceptual Model:

The new QOU Student Portal emphasizes instant information accessibility for the users. Rather than having to navigate several screens, all the vital information is now presented on one main screen, including:

- Daily Schedule

- Updated Grades

- Upcoming Tests/Assignments

- Course Information

The model prioritizes usability and reduces cognitive load, allowing students to complete tasks efficiently and with minimal navigation.

2. Persona-Based Design Considerations

Primary Persona: Mohammad Ahmad

Goals:

1. Check all grade publication statuses without opening individual course panels.
2. View weekly schedule without calculating odd/even weeks.

Frustrations addressed:

- Odd/Even week confusion
- Invisible individual assignment grades
- Unclear zero grades

Secondary Persona: Sujood

Goals:

1. Access grades, unread messages, and pending actions in under 3 minutes.
2. Quickly view and manage academic information relevant to her courses.

3. Complete tasks reliably within the portal without needing to visit the university.

Frustrations addressed:

Slow or unreliable administrative processes.

Messages hidden in submenus.

Unclear status of grades or exam reviews.

Some portal services appear to exist but are not available or functional when used.

3. Norman's Principles

The redesign of the QOU Student Portal applies Norman's six design principles to make the system easier to understand, faster to use, and less error-prone.

3.1 Affordances: The design of the interface provides the intended functions for the elements by their appearance and labeling. For instance, the labels on "View Grades", "Open Schedule", and "Read Message" buttons inform the student about what can be done through the buttons. It helps users understand what they can do without any additional explanations.

3.2 Signifiers: Clear visual and textual signifiers are used throughout the system to direct users to the next steps. For example, a red dot labeled "New" shows newly published grades or messages that have not been viewed yet. A greyed out button means that performing some actions such as reviewing exams is currently impossible.

3.3 Mapping: The design of the new portal corresponds to the student thinking style in respect to task completion. First of all, the most frequently used data appears on the portal dashboard: system status summary, schedule, grades, messages, course services and Exams, it allows answering the natural student question "What should I know today?" instead of making a student find required information in different sections of the portal

3.4 Feedback: Feedback provided by the portal is direct and appropriate. If no grade is available, the system displays "Not yet published" instead of leaving the field blank or showing zero. This prevents students from confusing an unpublished grade with a real grade of zero. Also, when an exam review is not available, the system clearly shows that the review period is closed instead of allowing the student to open a page that only displays an error message.

3.5 Constraints: Constraints are used in the redesigned portal to reduce mistakes and prevent students from misunderstanding schedule and grade information. In the weekly schedule, the system does not require students to calculate whether the current week is odd or even. Instead, it automatically displays the schedule for the current week and provides an option to view the schedule for any selected week. This keeps the odd/even logic inside the system rather than placing the burden on the student. In the grades section, the number 0 is displayed only when it is a real published grade. If the grade has not been released yet, the system shows "Not yet published." These constraints help prevent wrong assumptions about classes, grades, and portal information.

3.6 Conceptual Model: It is essential to mention that the newly designed portal has the conceptual model: a dashboard where all necessary updates are gathered together. Users will be able to realize that their dashboard serves as the first tool that should be used to receive information about schedules, grades, messages, tests, assignments, and other useful material. Such a model is helpful for the project personas, as Mohammad needs immediate access to grades and schedules, whereas Sujood is to finish her portal task quickly.

4. Nielsen's Heuristics

H1. Visibility of System Status: The revised portal structure makes sure that the students can observe the status of the most important subsystem all together from one summary card. Also they know the status of their grades or the exam review. There is a clear indication for each course grade whether it is "New grade published", "Viewed" or "Not yet published". Likewise, in case an exam review is not available, there is a clear indicator that the review process has been closed or not opened yet. Thus, no confusion is left to the students regarding the status of any grades or exam reviews.

H2. Match Between System and the Real World: In the redesigned portal, the user-friendly language is used instead of using technical terms and jargon. In particular, the headings on the schedule page will be something like "Classes for this week" instead of asking users to figure out whether it is an even or an odd week. The term "Not yet published" is used in place of blank cells in the case of the grades report.

H3. User Control and Freedom: The redesigned portal gives students clear control over their navigation without forcing them through unnecessary steps. From the dashboard, students can open the grades section, weekly schedule, exams, or assignments, then return easily to the main dashboard. The schedule view also allows students to check the current week or select another week when needed. This helps students move between important academic information without getting lost inside long menus or repeated pages.

H4. Consistency and Standards: Consistent layouts, wording, and interaction principles have been employed in the redesigned portal. The same card designs and statuses have been applied to grades, messages, exams, and assignments. There would be no need for students to wonder about the meaning of different words or visual styles, as defined by Nielsen.

H5. Error Prevention: Errors are avoided from the very beginning through the prevention approach. Links to unavailable exams for revision will be turned off and not activated; unpublishable grades are clearly marked and not represented as zeros. Additionally, the calendar does not imply the need for manual calculation of the odd/even weeks' numbers. This way, less errors regarding missed lessons, misunderstandings related to grades, and unnecessary visits to the faculty can occur.

H6. Recognition Rather than recall: With the portal's interface redesign, the load of users' memory is decreased. The students do not need to think what week number they are studying at now, whether it is odd or even. In addition, there is no need for them to recollect which subject might have some new message or grade.

H7. Flexibility and Efficiency of Use: The new dashboard provides adaptability and efficiency because it gives the ability to students to review the significant changes before accessing each section. The status indicators in the navigation menu indicate the current status of the grades and exams sections in red, green, or grey according to whether there is something new, viewed earlier, or nothing new in the section. In this way, they are aware of when there is something new, so they do not waste additional time accessing the same sections. Thus, both Mohammad and Sujood can access relevant information on their phones easily and within a few clicks.

H8. Aesthetic and minimalist design: Unlike the old portal that is characterized by too many unnecessary columns and information-filled tables, this redesigned interface utilizes simple cards and labels. The task-related information only is presented. For example, in the schedule tab, only the relevant columns are displayed such as class name, time, type, and location while also preserving the functionality of viewing more details that were moved to a pop-ups as a sort of layering information based on their relevance.

H9. Help users with errors: Since the redesigned portal focuses on preventing errors before they happen, this heuristic is addressed only when an error or unavailable action still occurs. For example, if a student tries to access an unavailable service, the system should display a clear message explaining that the service is currently not available, instead of showing a blank page or a technical error.

H10. Help and Documentation: Help information is provided if necessary in the form of concise tips. For instance, "Not yet published" could be further explained by a tooltip, as well as why a certain "exam review" button is greyed out. This meets the requirements of Nielsen who suggests that the help information be task-related, concise and easily accessible.

5. Cognitive Load Considerations

- Students do not need to calculate odd/even weeks manually because the system displays the current weekly schedule automatically.
- Students can check grade status from one overview screen without opening each course separately.
- Schedule and grade pages avoid displaying unnecessary information that does not help the student complete the task, least relevant information were moved to pop-ups.
- Cards, status indicators, and clear labels such as "New," "Viewed," and "Not yet published" reduce the mental effort needed to understand portal information.

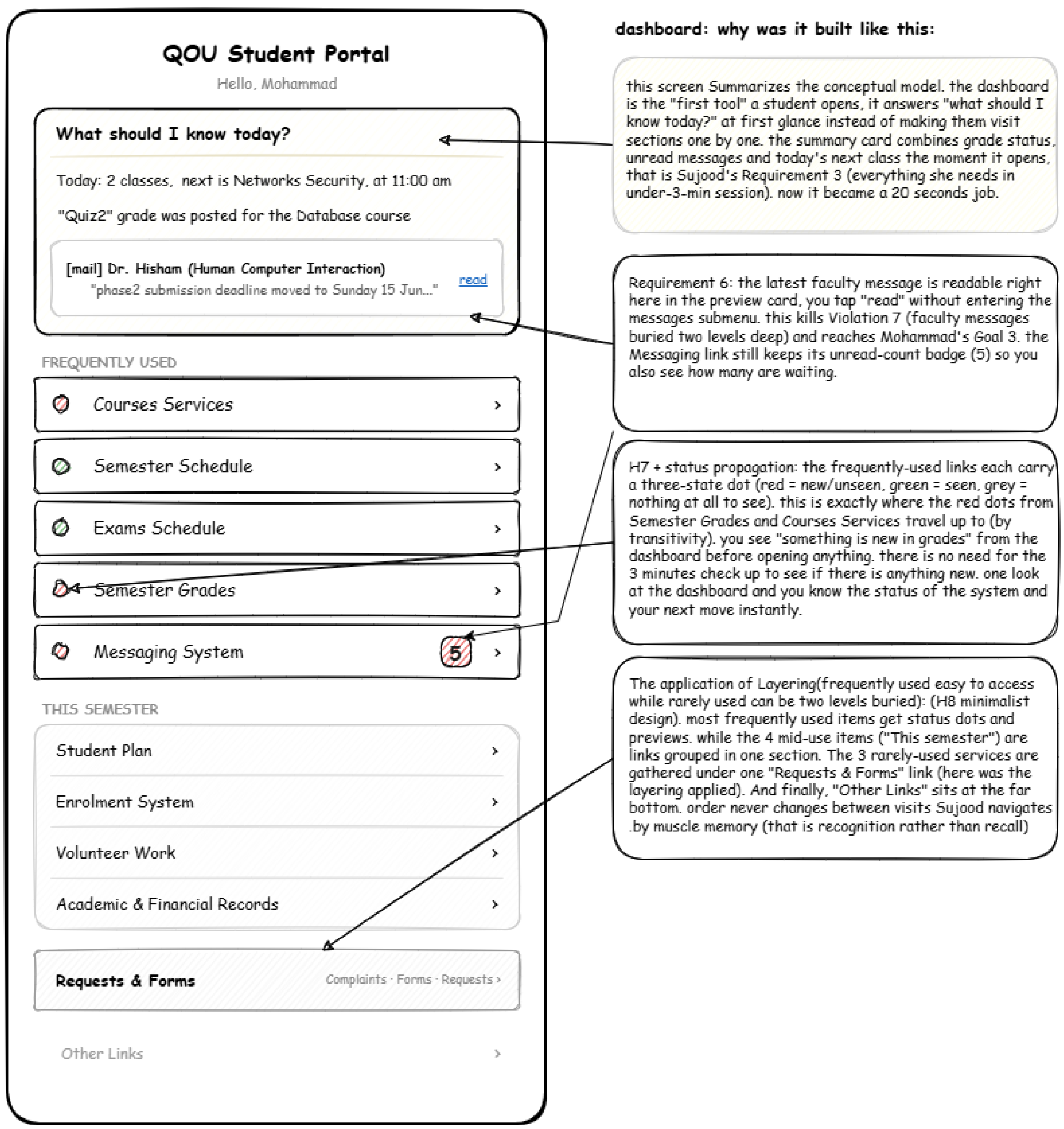
6. Accessibility

- - Full keyboard navigation support.
  - Screen reader compatible text labels.
  - WCAG 2.2 compliant color contrast.

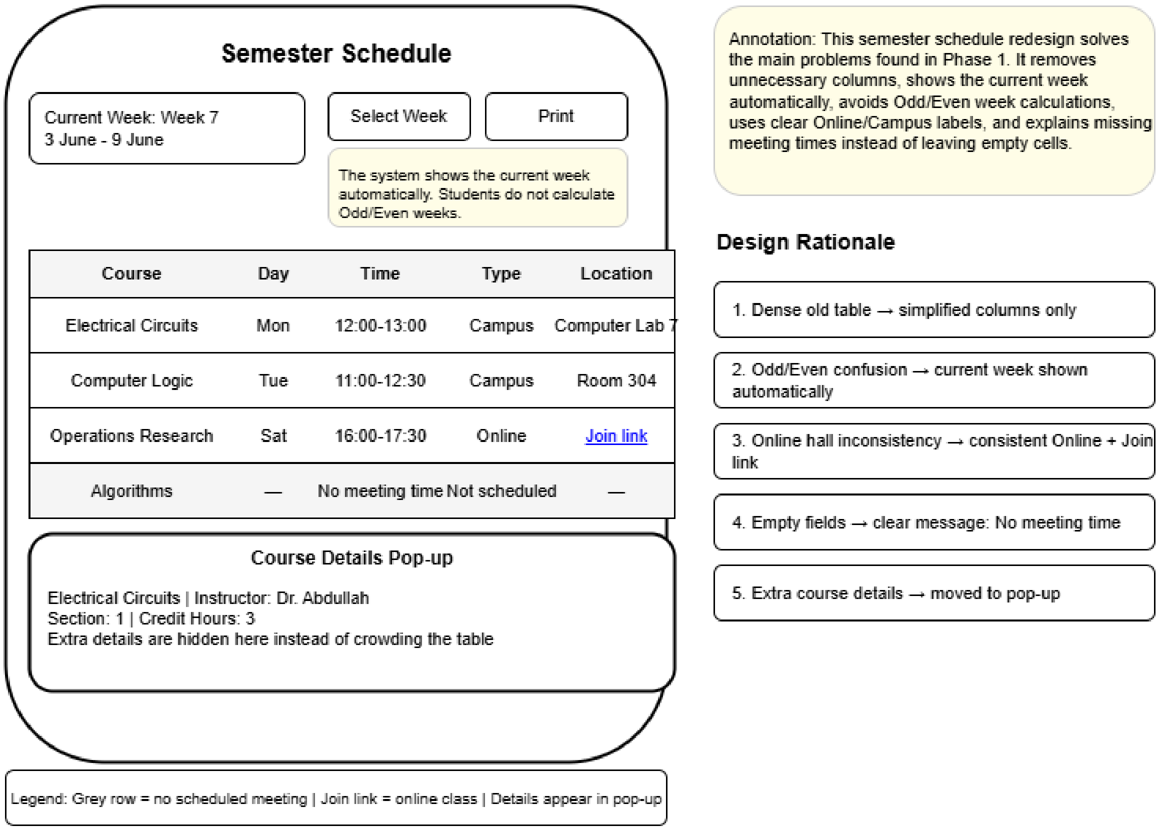
-Status changes are not communicated by color alone. Each color indicator is paired with a clear text label and icon, such as “New,” “Viewed,” or “Not available,” so color-blind users can still understand the status.

Low-fidelity wireframes:

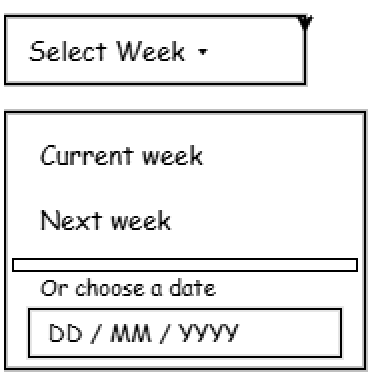
Dashboard Wireframe:-



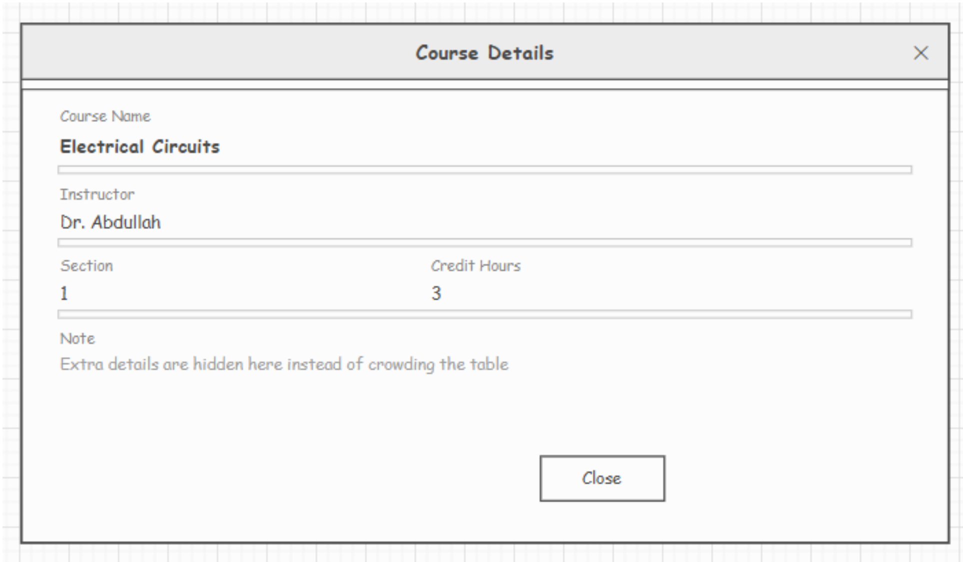
Semester Schedule Wireframe:



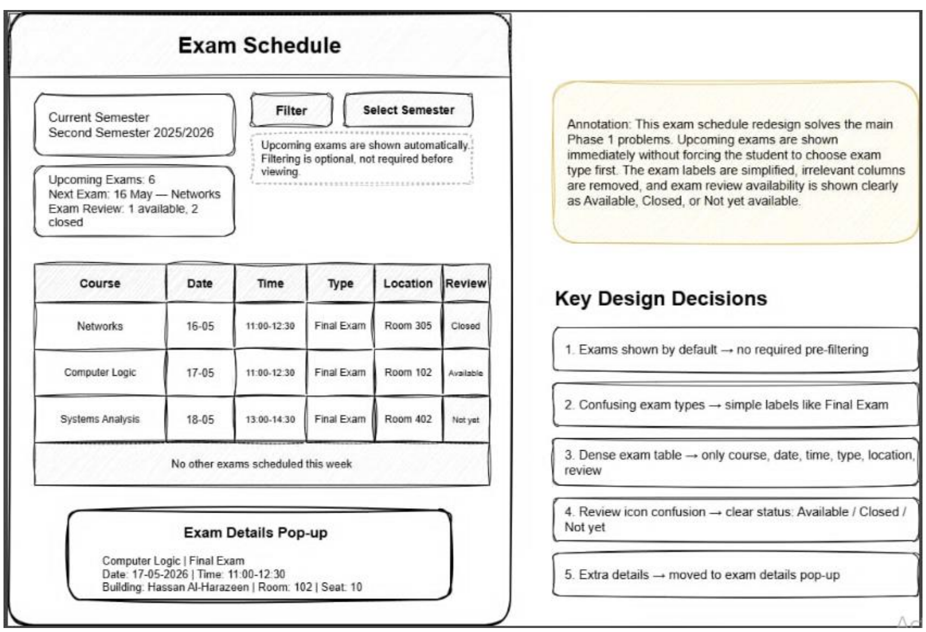
Select weeks drop down:



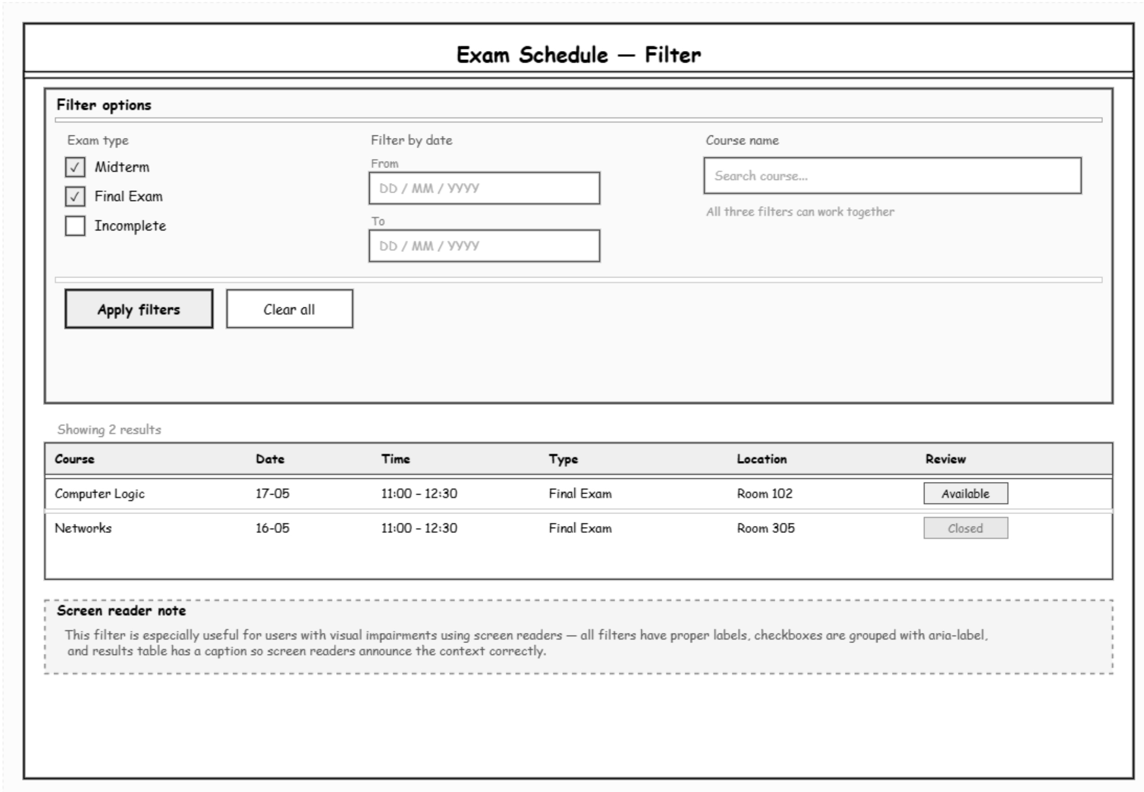
Course Details pop up Wireframe:



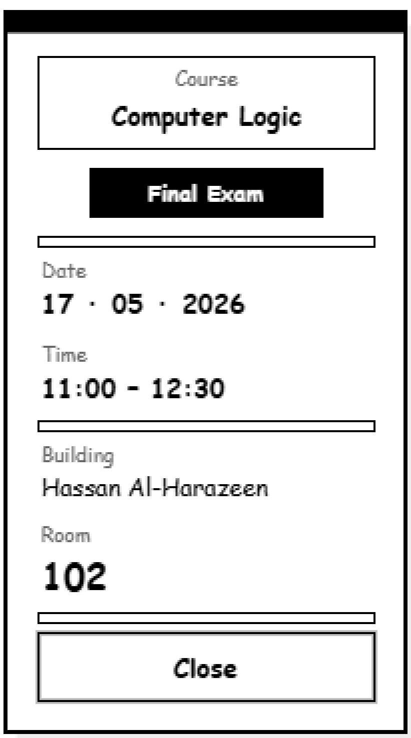
Exam Schedule Wireframe:



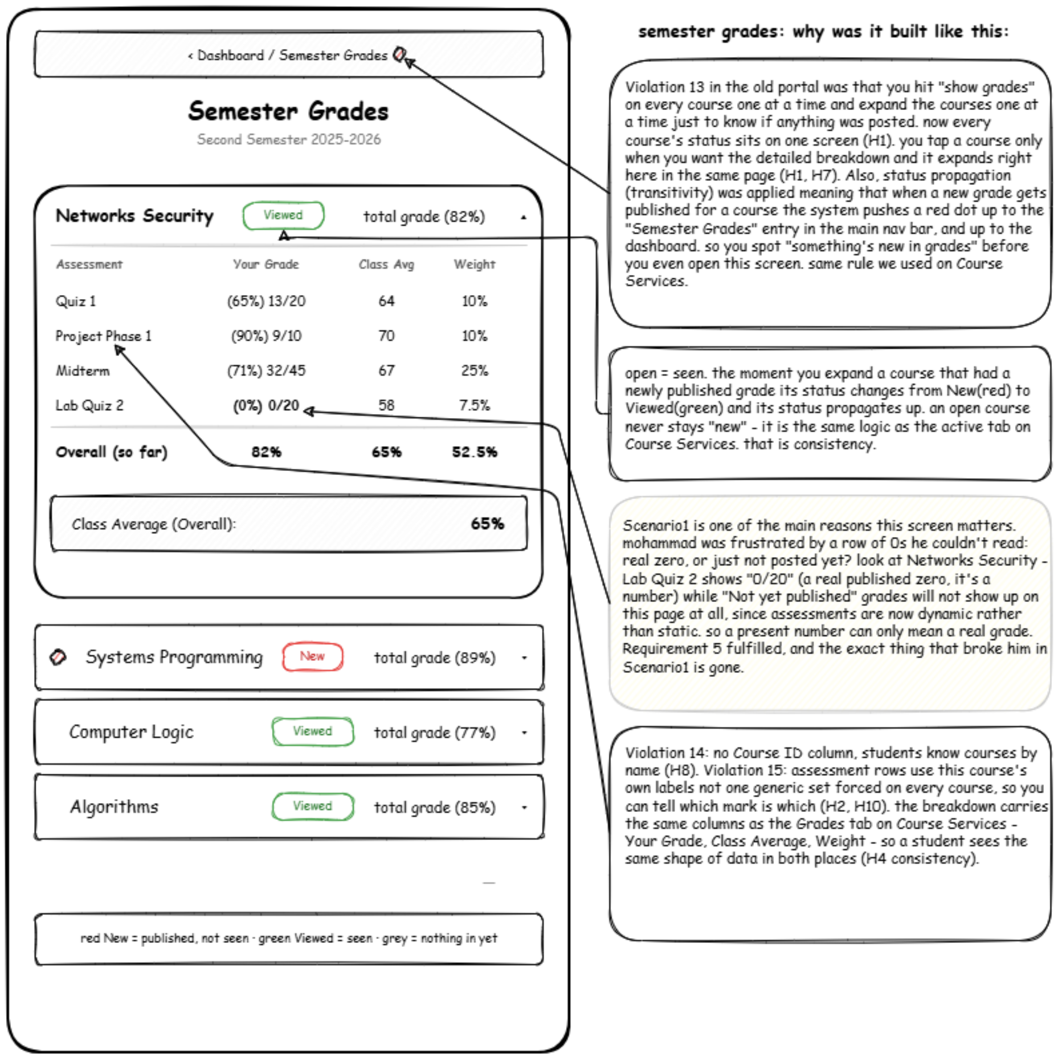
Filtering Wireframe:



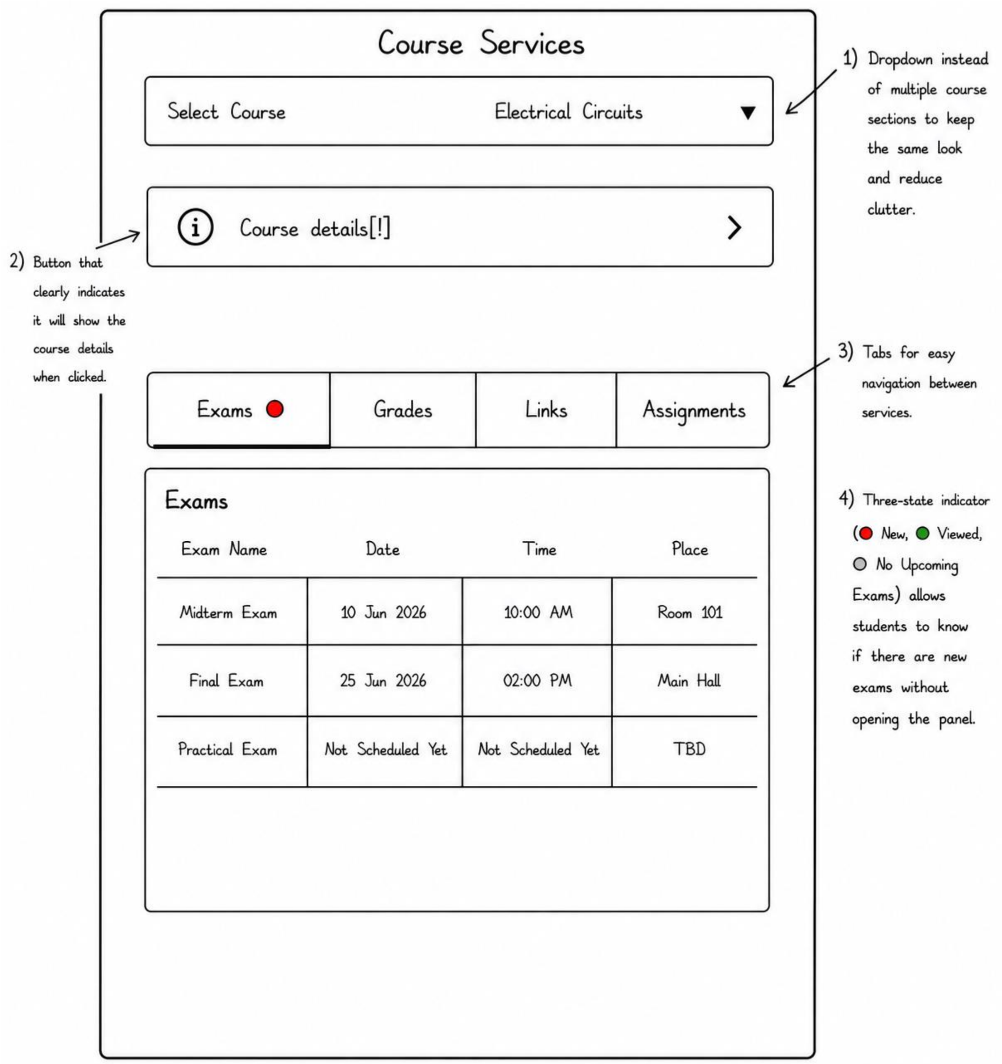
Exam Details pop-up Wireframe:



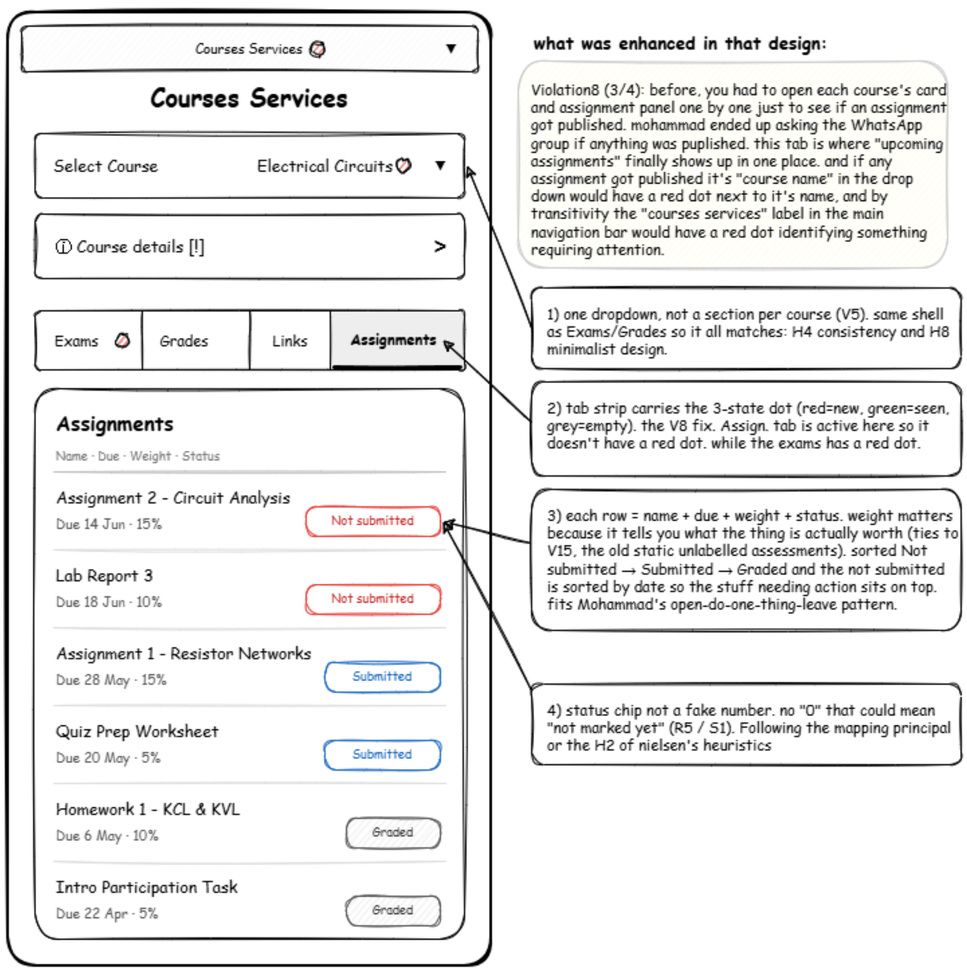
Semester Grades Wireframe:



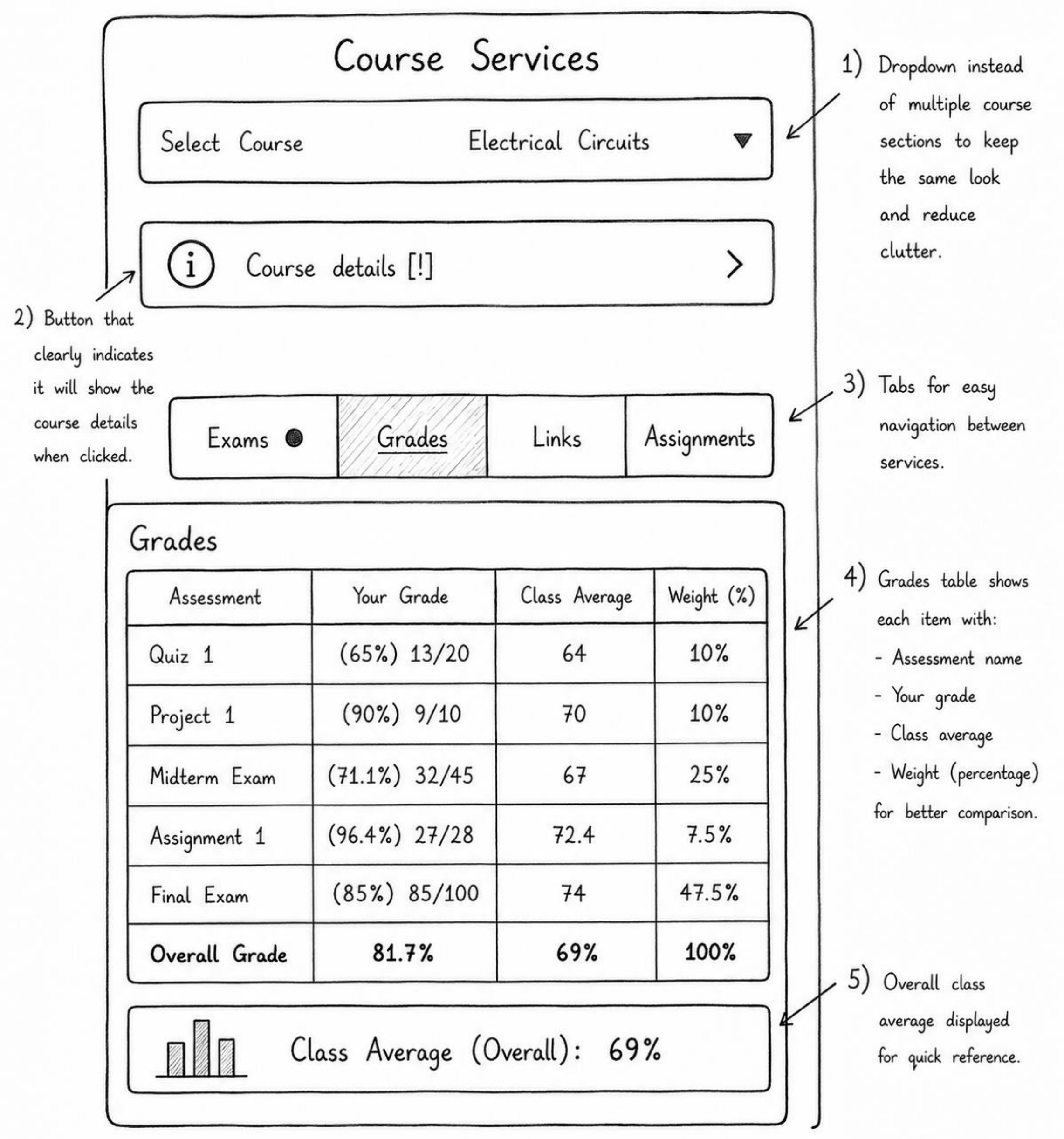
Course Services (Default appearance opened on the Exams service):



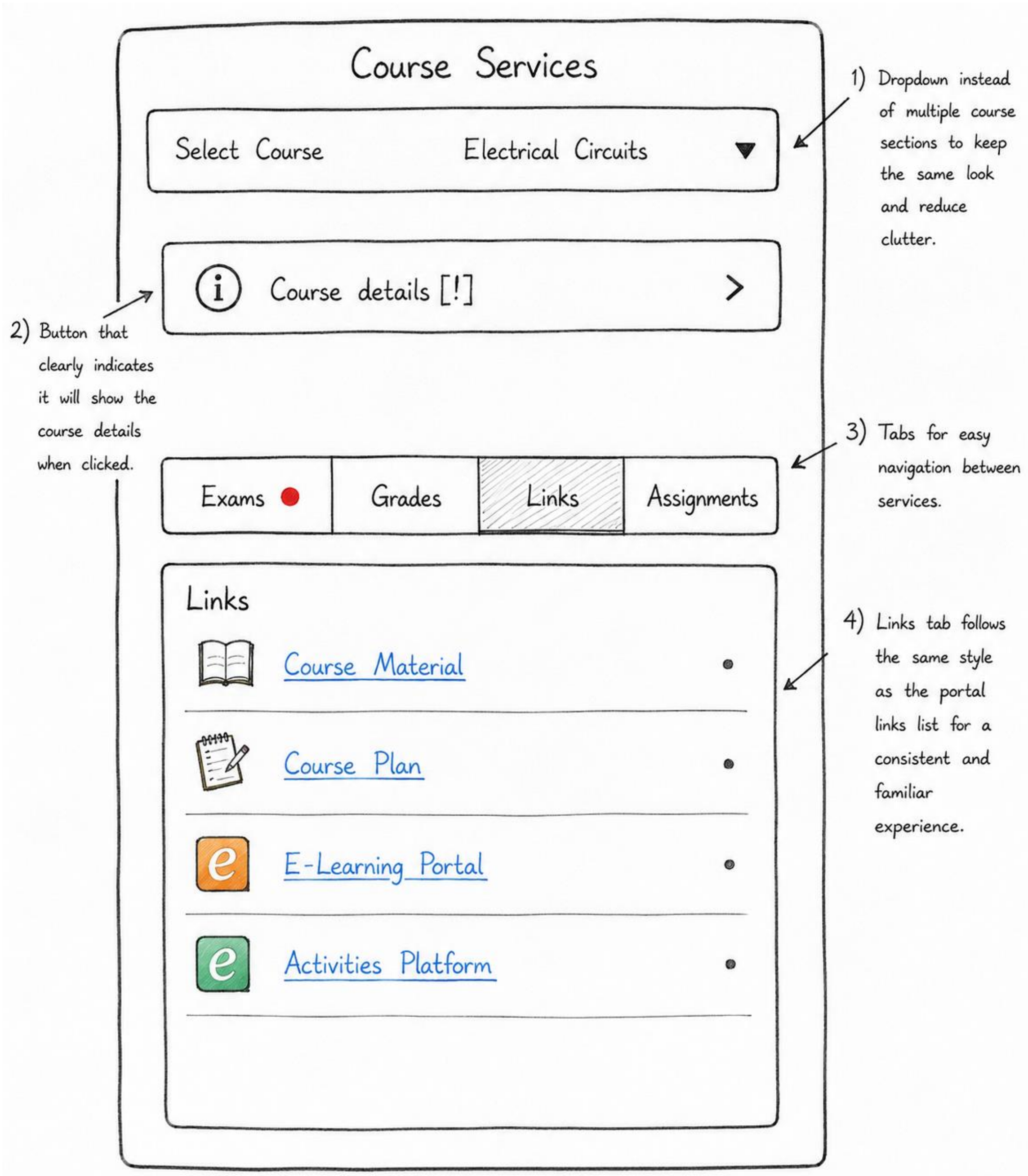
Course Services (Assignments service):



Course Services (Grades Service):

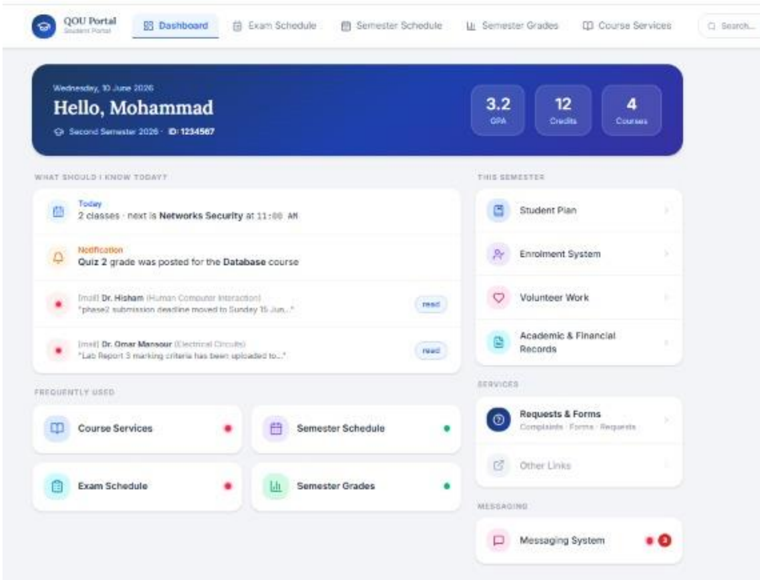


Course Services (Links Service):

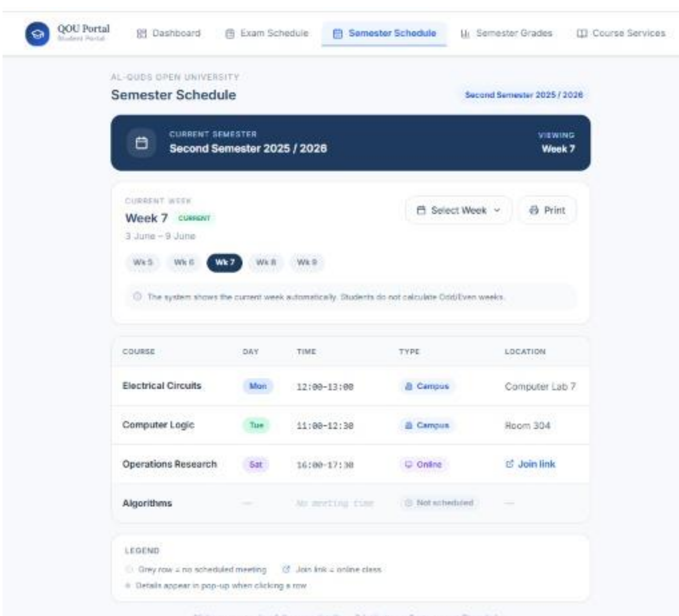


Interactive prototype:

Dashboard Wireframe:-

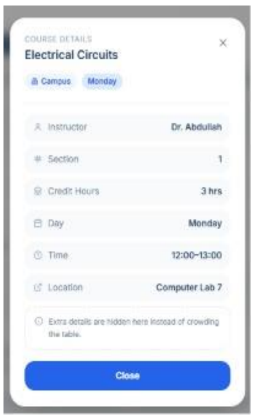


Semester Schedule Wireframe:

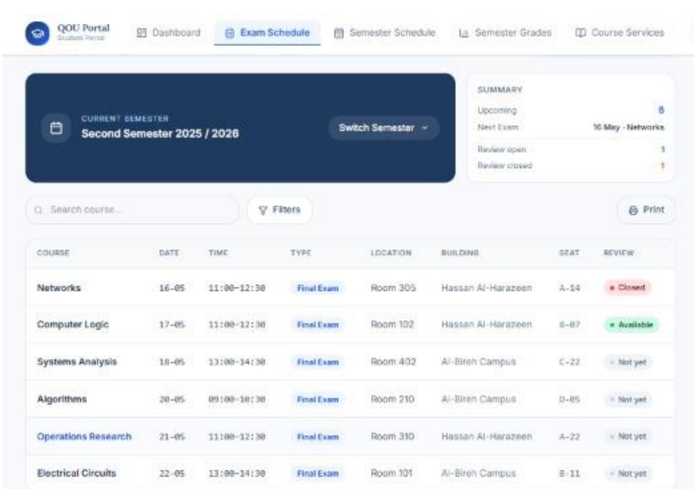


Select weeks drop down:

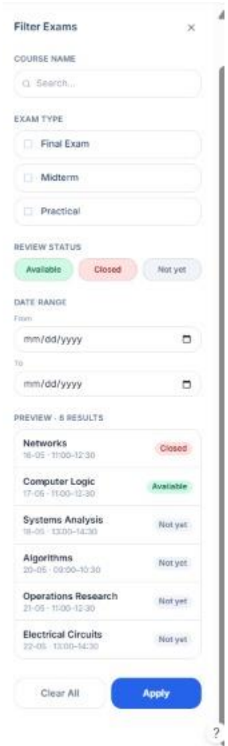
Course Details pop up Wireframe:



Exam Schedule Wireframe:



Filtering Wireframe:



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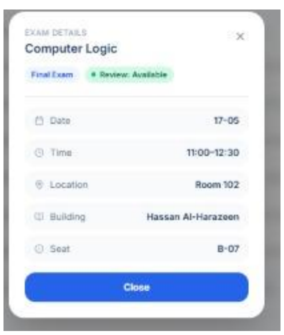
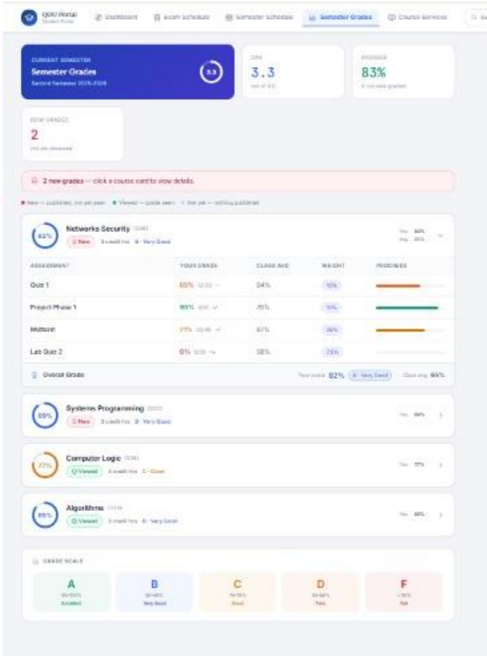


Semester Grades Wireframe:



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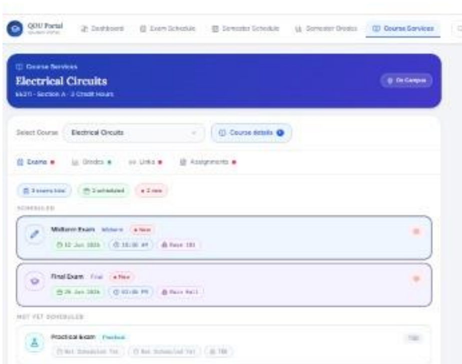

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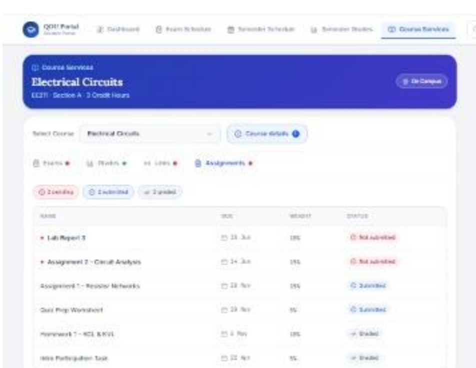

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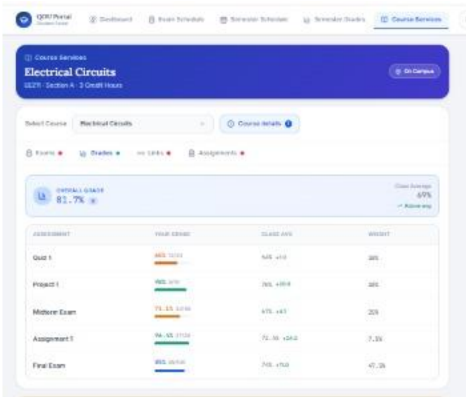

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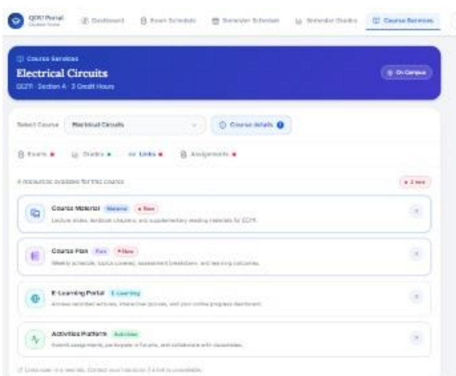

Link:

[**https://www.figma.com/make/bhCkhATEMJ9kgRO3nGhI5h/Exam-Schedule-Portal-Design?p=f&t=9g35n9JxpDr4wrrL-0**](https://www.figma.com/make/bhCkhATEMJ9kgRO3nGhI5h/Exam-Schedule-Portal-Design?p=f&t=9g35n9JxpDr4wrrL-0)